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CS606-Compiler Construction
Update MCQ'S Mid Term
By Vu Topper RM

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Rizwan Manzoor
Vu Topper RM



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A tool that translates, say, Urdu into English could be labeled as a_____.

A. Scanner
B. Parser
C. Compiler
D. Lexer

The front end of a compiler consists of _____ module(s).

A. Four
B. Three
C. Two
D. One

What is the typical output of a lexical analyzer?

A. The stream of tokens or lexemes
B. The intermediate code representation
C. The final executable program
D. The abstract syntax tree (AST)

While converting an NFA into a corresponding DFA, the start state of the DFA will be the _____ of the start state of the NFA.

A. α-closure
B. ε-closure
C. Δ-closure

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D. ϕ-closure

The acceptance of an NFA for a given string is achieved if it can get in a _____ state.

A. Error
B. Start
C. Dead
D. Final

What is the resultant value of following expression?
Expression: 7 + 2 * (9 / 3) - 1

A. 26
B. 10
C. 28
D. 12

In a context-free grammar, the non-terminal symbol that represents the starting point of the language is called a/an _____.

A. Start
B. End
C. Unique
D. Unit

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For each production rule A → α in an LL(1) parsing table, if there is another production rule A → β, then FIRST(α) ∩ FIRST(β) = ?

A. β
B. Ø
C. α
D. γ

What is the significance of the term "context-free" in a context-free grammar?

A. It implies that the grammar is free from any restrictions or limitations.
B. It signifies that the rewriting rules are independent of the surrounding symbols.
C. It indicates that the grammar can generate all possible languages.
D. It specifies that the grammar is designed to handle complex contextual dependencies.

The non-terminal symbols encountered during parsing are stored in a:

A. Linked list
B. Binary tree
C. Queue
D. Stack

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It is said that the syntax of C, C++ and Java is derived heavily from _____.

A. Algol-60
B. Algol-70
C. Ada-70
D. Ada-60

LR parsers CAN NOT handle left-recursive grammars.

A. True
B. False

The production rule "S → s S | ε " represents a Left-Recursive Grammar.

A. No
B. Yes

If S → c S d a, then FIRST (S) = ?

A. { S }
B. { d }
C. { c }
D. { a }

In recursive descent parser, the term "descent" refers to the direction in which the parse tree is built.

A. False

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B. True

If S → a T b and T → c then FIRST (T) = ?

A. { c }
B. { a }
C. { $ }
D. { ε }

For the grammar "S → a c T | a U", it is very easy to predict because two productions start with 'a'.

A. True
B. False

In a recursive descent parser, Factor is defined as a C++ class that contains following two public methods;

A. A destructor and an abstract method
B. A destructor and a friend method
C. A constructor and a virtual method
D. A constructor and a simple method

The statement "construct the root node of the parse tree" is considered as a first step in a top-down parsing algorithm.

A. True
B. False

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In an LL(1) grammar, there should be NO conflicts (i.e., multiple entries in a cell) in the parsing table.

A. True
B. False

If S → S T and T → t, then FOLLOW (T) = ?

A. { $, S }
B. { $, t }
C. { ε, S }
D. { ε, t }

The LL(1) parsing table is used by the LL(1) parser to parse the input and generate a parse _____.

A. Stack
B. Queue
C. Array
D. Tree

Grammar with the _____ property is called a predictive grammar.

A. RR(1)
B. RL(1)
C. LL(1)
D. LR(1)

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In shift-reduce parsing, Shift action moves ► one place to the _____ which shifts a terminal to the left string.

A. Top
B. Left
C. Right **Page 71**
D. Bottom

In a recursive descent parser, Factor can be defined as a C++ class which should inherit from _____ class.

A. Terminal
B. Non-terminal
C. Token
D. Node

The convention in most programming languages is to match the else with the _____ if.

A. Most Upcoming
B. Very Last
C. Very First
D. Most Recent

In LR grammar, the R stands for _____.

A. Right most derivation
B. Right hand side of grammar

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C. Right to left scan
D. Right shift

If find ε in FIRST(α), the production rule "A → α" should be added to the LL(1) parsing table entry M[A, b] for each terminal 'b' in:

A. FIRST(A)
B. FIRST(B)
C. FOLLOW(A)
D. FOLLOW(B)

In a context-sensitive grammar, what does the term "context-sensitive" refer to?

A. The grammar's ability to recognize patterns without considering context.
B. The grammar's ability to rewrite non-terminals regardless of context.
C. The grammar's ability to generate strings based on surrounding context.
D. The grammar's ability to handle any number of terminals and non-terminals.

The main advantage of non-recursive predictive parsing is:

A. Avoidance of stack overflow errors
B. Simpler implementation

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C. Reduced memory usage
D. Faster parsing speed

For each _____, add production rule "A → α" to LL(1) parsing table entry M[A, a].

A. Non-terminal 'A' in FIRST(α).
B. Terminal 'a' in FIRST(α).
C. Non-terminal 'A' in FOLLOW(α).
D. Terminal 'a' in FOLLOW(α).

In the context of recursive descent parser in C++, what is the purpose of the tree variable in the Eprime class?

A. To store the intermediate abstract syntax tree.
B. To store the result of the isPresent() function.
C. To represent the root node of the abstract syntax tree.
D. To keep track of the current expression being parsed.

In the context of recursive descent parser in C++, what is the return type of the isPresent() function in the Eprime class?

A. double
B. int
C. float
D. bool

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After performing left factoring on the grammar "S → a X | a", the resultant grammar will be;

A. S → X a, S'→ S | ε
B. S → a X, S'→ S | ε
C. S → a S', S'→ X | ε
D. S → S' a, S'→ X | ε

If E → T and T → a, then FOLLOW (T) = ?

A. { E }
B. { T }
C. { $ }
D. { a }

Identify a left recursive grammar from the following;

A. S → a S b c | ε
B. S → a b S c | ε
C. S → S a b c | ε
D. S → a b c S | ε

In the grammar "T → a b c d | a b d c", the _____ is a common prefix for non-terminal 'T'.

A. b d
B. a c
C. c d

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D. a b

Which of the following is an example of a context-sensitive grammar rule?

A. αAβ → αγβ
B. A ← α
C. αAβ ← αγβ
D. A → α

Which of the following production rules indicates a nullable non-terminal?

A. S → a
B. U → c
C. V → ε
D. T → b

In compiler construction, an LR parser is a type of:

A. Bottom-up parser
B. Recursive descent parser
C. Top-down parser
D. Non-recursive parser

In the reduce action, the shift-reduce parser applies an inverse production at the right end of the _____ string.

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A. Top
B. Left **Page 72**
C. Right
D. Bottom

In LR grammar, the L stands for _____.
A. Left most derivation
B. Left to right scan
C. Left hand side of grammar
D. Left Shift

In the grammar "A → a d c b | a d b c", the _____ is a common prefix for non-terminal 'A'.
A. b
B. a b c d
C. d c b
D. a d

Non-recursive predictive parsing is a technique used for:
A. Optimizing the performance of recursive functions.
B. Analyzing runtime behavior of programs.
C. Debugging software applications.
D. Analyzing the syntax of a programming language.

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In the context of recursive descent parser in C++, what does the "s → nextToken() != RPAREN" condition check?

A. If the next token is not a right parenthesis.
B. If the next token is an identifier or number.
C. If the next token is a left parenthesis.
D. If the next token is an arithmetic operator.

The parsing completion criteria in non-recursive predictive parsing include:

A. Non-empty stack and remaining input tokens
B. Empty stack and remaining input tokens
C. Empty stack and consumed input tokens
D. Non-empty stack and consumed input tokens

Which of the following is a valid production rule in a context-free grammar?

A. B ← A + C
B. A + C → B
C. A + C ← B
D. B → A + C

Which of the following is an example of a context-free grammar rule?

A. A ← α
B. αAβ ← αγβ

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C. A → α

D. αAβ → αγβ

In a recursive descent parser, Expr can be defined as a C++ class which should inherit from _____ class.

A. Token

B. Node

C. Terminal

D. Non-terminal

The rule "S → ε" says that the symbol S CAN NOT be rewritten as an empty string.

A. True

B. False

If FIRST(α) = {ε} and _____, then add production rule "A → α" to LL(1) parsing table entry M[A, $].

A. FOLLOW(α) = {a}

B. FOLLOW(α) = {$}

C. FOLLOW(α) = {A}

D. FOLLOW(α) = {ε}

Top-Down parser is also called a/an _____ parser.

A. Predictive

B. LR(0)

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C. Bottom-Up
D. Shift-Reduce

In the production rule "A ⟶ b C d", what symbol appears on the left side?

A. C
B. b
C. d
D. A

In arithmetic, addition has _____ priority over multiplication.

A. Lower
B. Higher
C. Equal
D. Same

In the production rule "A ⟶ b C D", the symbol D appears on the _____ side of the production rule.

A. Left
B. Right
C. Bottom
D. Top

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In the context of LL(1) parser, if top symbol of stack 'X' and current input symbol 'a' are equal to end-of-input marker '$' (i.e., X = a = $) then the parser;

A. Pops 'X' from Stack.
B. Generates an error message.
C. Announces successful completion.
D. Pushes 'a' into Stack

If S → x y z a, then FIRST (S) = ?

A. { y }
B. { a }
C. { z }
D. { x }

The formal definition of FIRST set is:

A. FIRST(X) = {b | X →* ba} ∪ {ε | X →* ε}
B. FIRST(X) = {b | X →* ba} ⊂ {ε | X →* ε}
C. FIRST(X) = {b | X →* ba} ∩ {ε | X →* ε}
D. FIRST(X) = {b | X →* ba} ⊃ {ε | X →* ε}

If S ⟶ T U c, T → t and U → ε, then FOLLOW (T) = ?

A. { ε }
B. { c }
C. { t }

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D. { $ }

The front end maps legal _____into an Intermediate Representation (IR).

A. Object code

B. Machine code

C. Linker

D. Source code ok

_____ is the regular expression to represent all binary strings in which last bit is 0. A. (0|1)*1

B. (0|1)*0 ok

C. 0(0|1)*

D. 1(0|1)*

Is it true that no algorithm exists for an "ideal translation"?

A. Yes

B. No

_____is not the role of Run-Time System?

A. Garbage collection

B. Run-time type checking

C. Syntax analysis ok

D. Memory management

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_____ is a compiler used in Java.

A. JavaD

B. JavaB

C. JavaE

D. JavaC ok

_____ are generally ignored by the compiler.

A. Strings

B. Integers

C. White Spaces ok

D. Keywords

Which of the following strings belong to the language specified by this regular expression: (abb)*a

A. bba

B. a

C. aaa ok

D. ba

In the context of token role in a lexical analyzer, identify a "Symbol" from the following;

A. If

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B. = = **ok**
C. 12
D. y11

Which of the following commands is used to generate a scanner executable file in Windows? A. g++ –o main.o lex.exe lex.o
B. g++ –o lex.o main.o lex.exe
C. g++ –o lex.o lex.exe main.o
D. g++ –o lex.exe lex.o main.o **ok**

An application that converts a Word file to PDF can be called as a/an _____.
A. Debugger
B. Translator
C. Linker
D. **Processor**

In C++, _____ refers to spaces, tabs and newlines etc.
A. Whitespace
B. Struct
C. Character
D. String

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If an NFA has a single state b, which has a self-loop on input 0, then move (b, 0) = ?

A. b

B. 0

C. ϕ

D. ε

How many tokens are there in the following statement? a = b+++c--d+e ;

A. 10

B. 11

C. 12

D. 13

_____ architecture made the job of the compiler harder when it came to generate efficient machine code.

A. CISC

B. VLIW **ok**

C. RISC

D. EPIC

If a production is A → XYZ, then parse tree will have _____ terminal/s.

A. One

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B. Three
C. Four
D. **Two**

In a transition table, cells of the table contain the _____ state.
A. Next State
B. Previous State
C. Accept state
D. Reject State

Which component in a CPU provides high-speed access to operands?
A. ALU
B. Cache
C. Memory
D. Register

_____ is a process used to reduce the number of states in a DFA while preserving the language it recognizes.
A. DFA Minimization ok
B. DFA Maximization
C. NFA Maximization
D. NFA Minimization

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If an NFA has a single state c, then ε-closure (c) = ? A.
ε
B 21
C. ϕ 21
D 32

The back end maps _____ into target Machine code.
A. Intermediate Representation (IR)
B. Linker
C. Machine code
D. Object code

In C++, the symbol for "bitwise XOR" operator is _____.
A. |
B. &
C. $
D. ^

The _____ code typically runs as a process in an Operating System Environment.
A. Executable
B. Object
C. Machine

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D. Source

Which of the following is an example of a valid identifier?

A. _myVariable

B. 123variable

C. variable name

D. 2ndVariable

For building a/an _____, the tokens are encoded using regular expressions. A. Semantic Analyzer

B. Lexical Analyzer

C. Optimistic Analyzer

D. Syntactical Analyzer

In Flex, what does the regular expression [a-zA-Z][a-zA-Z0-9]* match?

A. Symbols

B. Integers

C. Whitespaces

D. Identifiers **ok**

Code optimization is performed in the _____ of a compiler. A. Back-end

B. Middle-end

C. Front-end

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D. Last-end

The machine code generated by a compiler must execute precisely the same computation as the source code.

A. True

B. False

In subset construction algorithm, _____ is a set of states which are reachable from state T on ε-transitions.

A. follow (T, α)

B. first (T)

C. move (T, α)

D. ε-closure (T)

In a parse tree, the root is labeled by the _____ symbol.

A. Next

B. Previous

C. Start ok

D. Last

A compiler converts a/an _____ language code into a lowlevel machine code. A. Low-level

B. Bottom-level

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C. High-level
D. Medium-level

The primary goal of the _____ of a compiler is to reduce running time of the compiled code.
A. Middle-end
B. Start-end
C. Back-end **ok**
D. Front-end

In C++, the symbol for "Bitwise Complement" operator is _____.
A. ~
B. |
C. &
D. $

Most of the modern compilers contain _____ stages.
A. Three
B. Four
C. Two
D. One

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How many tokens are there in the following statement? for (int i = 0 ; i < 10 ; i++) { /*body of loop*/ }

A. 17

B. 16

C. 18

D. 15

In Flex, which section of a specification file contains "Token definitions and actions"? A. Fourth Section

B. Third Section

C. First Section

D. Second Section

The regular expression that allows all upper-case alphabets is _____.

A. [A/Z]

B. [A*Z]

C. [A-Z]

D. [A+Z]

Interpreters can be used for scripting languages, which are designed to automate tasks and provide a high level of abstraction.

A. True ok

B. False

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The parse tree contains a lot of unneeded information. Therefore, compilers often use a/an _____.

A. Balanced Tree

B. AVL Tree

C. Binary Search Tree

D. Abstract Syntax Tree

If an NFA has two states (a & b) and a transition from a to b on input 1, then move (a, 1) = ?

A. a

B. ε

C. b

D. 0

Instruction Selection is an important part of _____ module of a compiler. A. Front-end

B. Start-end

C. Back-end

D. Middle-end

Which approach is most frequently used to handle the complexity of translating programming languages?

A. Single-Pass translation

B. Recursive translation

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C. Multi-Pass translation ok

D. Automated translation

In subset construction algorithm, _____ is a set of NFA states to which there is a transition on input α from some NFA state S in set of states T. A. first (T)

B. follow (T, α)

C. ε-closure (T)

D. move (T, α)

What is the output of a back-end module of a compiler?

A. Source Code

B. IR

C. Token

D. Machine Code

In a parse tree, each interior node is labeled by a _____. A. Root

B. Non-Terminal

C. Terminal

D. Token

In Flex specification file different sections are separated by _____ .

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A. &&
B. //
C. ##
D. %%

Front-end of a compiler takes_____ as input.
A. Binary code
B. Object code
C. Machine code
D. Source code

“ ”, “ab” , “abab”... are the strings of a language denoted by following Regular Expression:
A. a
B. ab
C. (ab)* ok
D. alb

What does GCC stands for?
A. GNU Compiler Codes
B. GNU Compiler Collections
C. GNU Computer Codes
D. GNU Computer Collections

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_____ translates information from one representation to another.

A. Compiler
B. Scanner
C. Debugger
D. Translator

Compared to compilers, interpreters are best for _____ languages. A. Object Oriented
B. Functional
C. Scripting
D. Procedural

In C++, the symbol for "Shift Right" operator is _____.
A. >
B. >> **ok**
C. >>>>
D. >>>

Context-free syntax is specified with a grammar G = (S, N, T, P), where production rules are represented by _____.
A. P
B. T
C. N

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D. S

In the context of token role in a lexical analyzer, identify an "Identifier" from the following;

A. 12

B. = =

C. if

D. **y11**

_____avoids hardware stalls and interlocks.

A. Instruction scheduling

B. Instruction selection

C. Memory allocation

D. Register allocation

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